



Sri Lanka Institute of Information Technology

Devma Cake n' Party Management System

Sprint 0

Information Systems Project Management 2026

Project ID: ISE_WE_0101_30

Submitted by:

1. IT24100948 - Ashcharya S.H
2. IT24102134 - Kodikarage Y.N.K
3. IT24102100 - Mendis M.M.R
4. IT24102136 - De Silva D.H.C

Submitted to:

Buddhima Attanayaka

Date of submission
25 August 2026

Criteria	Marks allocated	Marks given
Problem Specification	5%	
Business Idea	5%	
Solution Outline	10%	
Key Benefits	10%	
Main Objectives	5%	
Sub Objectives	5%	
Project Plan	10%	
Product Vision statement	05%	
Epics	05%	
User Stories	10%	
Initial product Backlog	05%	
Definition of Done	05%	
Hardware & Software requirements	5%	
Budget	5%	
References	5%	
Formatting	5%	
Total Marks (100)	100%	

Team Members:

Student ID	Student Name	Role (Product Owner / Scrum Master / Team Member)
IT24100948	Ashcharya S.H	Product Owner
IT24102134	Kodikarage Y.N.K	Scrum Master
IT24102100	Mendis M.M.R	Team Member
IT24102136	De Silva D.H.C	Team Member

1. Introduction

1.1 Problem Specification

Devma Cake n' Party is a business that handles cake orders and party decoration products for its customers. Now Devma Cake n' Party handles its main tasks through manual and split communication channels such as WhatsApp, Facebook and telephone calls. There is no web-based system to manage customer information, products, inventory orders, payments and delivery activities.

This manual method makes it hard to keep customer and order records to watch product availability and stock levels and to manage cake requests and party decoration choices. It can also cause communication, manual mistakes, delays in processing orders and trouble keeping business information consistent.

Therefore Devma Cake n' Party needs a web-based Cake and Party Decoration Management System. That system will help Devma Cake n' Party manage information better cut down on work make day-, to-day operations smoother and give Devma Cake n' Party better control over its everyday activities.

1.2 Business idea

The proposed business idea is to develop a web-based Cake and Party Decoration Management System that centralizes the key operations of the business in a single platform. The system will support internal staff access management, product and inventory management, customer and cake order management, as well as order confirmation, payment, and delivery management.

The system will create business value by providing a centralized and organized way to manage operational information, reduce dependency on manual processes, improve information visibility and accuracy, and support more efficient day-to-day operations. It will also provide a structured platform for managing customer orders, monitoring product availability, and coordinating order-related activities from confirmation to completion.

1.3 Solution Outline

The proposed solution is a web-based Cake and Party Decoration Management System that brings all the business activities into one platform. The web-based Cake and Party Decoration Management System gives functions for customers for internal staff and for the business owner or administrator. Customers can look at the products that're available they can place orders, for cakes and they can choose party decoration items. At the time internal staff and the business owner will manage customers manage products manage inventory manage orders manage payments and manage delivery or collection activities.

The web-based Cake and Party Decoration Management System will run on a web application that connects to a database. The application will take care of the business processes in the application and business logic layer. These processes include User and Access Management, Cake and Party Decoration Product and Inventory Management, Customer and Cake Order Management Order Confirmation, Payment and Delivery Management and Management and Reporting. The central database will keep records of users and roles, product and inventory data, customer records order and cake requirement details, payment and order status data, delivery or collection records and business reports.

High-Level System Architecture

The system architecture that is being suggested has the following parts:

- **Users** – People such as Customers, Internal Staff and Business Owner or Admin who use the system based on what they need to do.
- **Web Application or User Interface** – This is the part that allows customers to look at products and order them and lets staff and the business owner handle the business tasks.
- **Application or Business Logic** – This part takes care of user and access control, managing products and inventory handling customers and orders confirming orders, managing payments and deliveries and creating reports.
- **Central Database** – This is where information, about users and their roles, products and inventory customer details orders and cake needs, payment and order status, delivery or pickup records and reports and business records are kept.

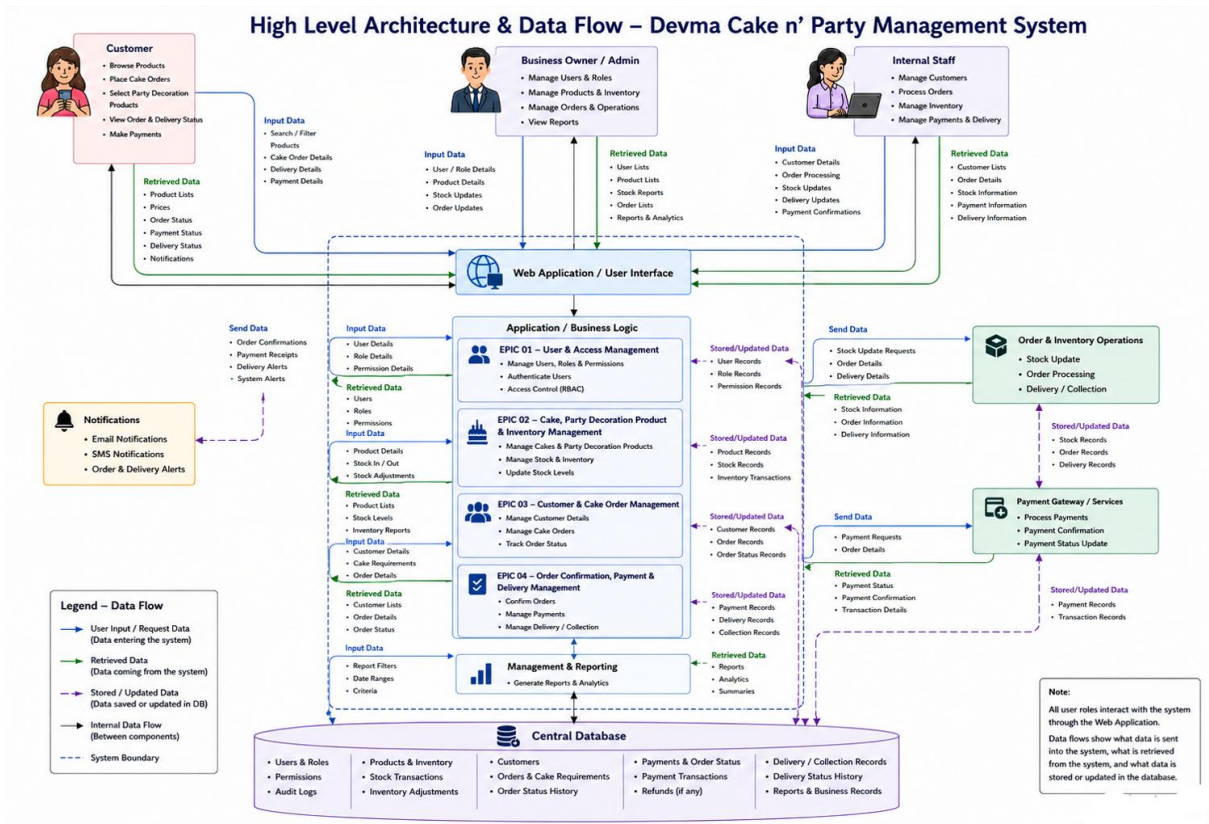


Figure 1: High-Level System Architecture of the Proposed Cake and Party Decoration Management System

1.4 Key Benefits

The proposed system is expected to provide the following benefits:

1. Reduced Manual Work

Automates. Centralizes routine business activities reducing time spent on manual data entry and record keeping.

2. Improved Data Accuracy

Reduces errors and inconsistencies in customer, product, inventory order and payment records.

3. Better Inventory Control

Provides updated product and stock information to improve availability monitoring and inventory management.

4. Efficient Order Management

Streamlines cake and party decoration order processing and improves visibility of order status and requirements.

5. Improved Customer Experience

Provides a platform for customers to browse products and place orders while improving order information accuracy.

6. Secure Staff Access

Role-based access ensures that staff can access the functions and information relevant, to their responsibilities.

7. Better Decision-Making

Management reports provide business information to support monitoring and operational decision-making.

8. Future Scalability

Provides a foundation that can support the business as its products, customers and operational activities grow.

2. Objectives

2.1 Main Objective

To develop a web-based Cake n' Party Management System that centralizes user access, product and inventory management, customer and cake order management, payment and delivery operations, and reporting, in order to reduce manual work, improve operational efficiency, and provide better customer service.

2.2. Sub Objectives

1. User & Access Management

To develop a centralized user and access management module that manages staff registration, roles, credentials, and access permissions, in order to improve security, reduce unauthorized access, and reduce manual verification work.

2. Cake, Party Decoration Product & Inventory Management

To develop a product and inventory management module that manages cake products, party decoration items, stock levels, and stock updates, in order to reduce stock errors, prevent stock shortages, and improve inventory control.

3. Customer & Cake Order Management

To develop a customer and cake order management module that records customer details, cake requirements, and order information, in order to reduce order errors, improve order tracking, and provide better customer service.

4. Order Confirmation, Payment & Delivery Management

To develop a module for order confirmation, payment, delivery, and collection management, in order to reduce delays, improve payment and delivery tracking, and ensure smooth order completion.

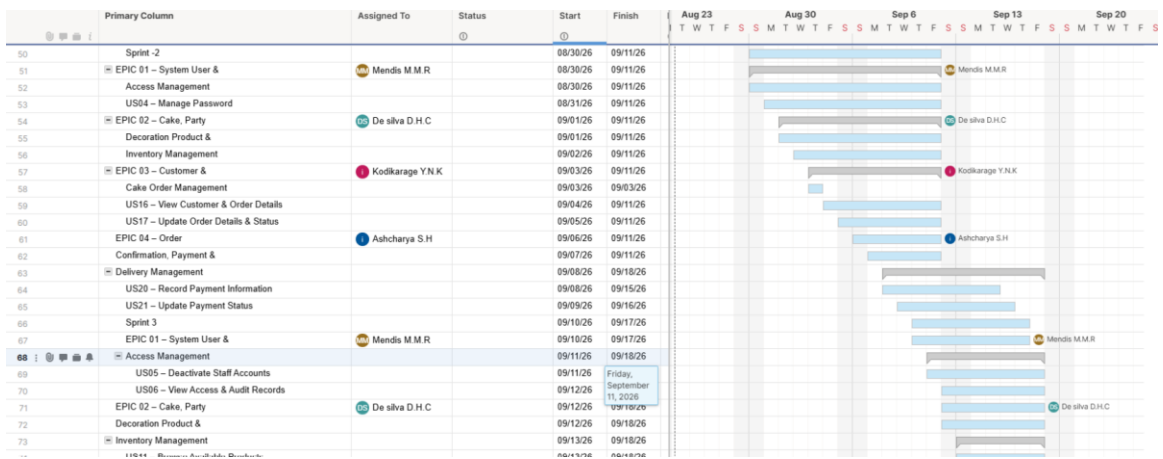
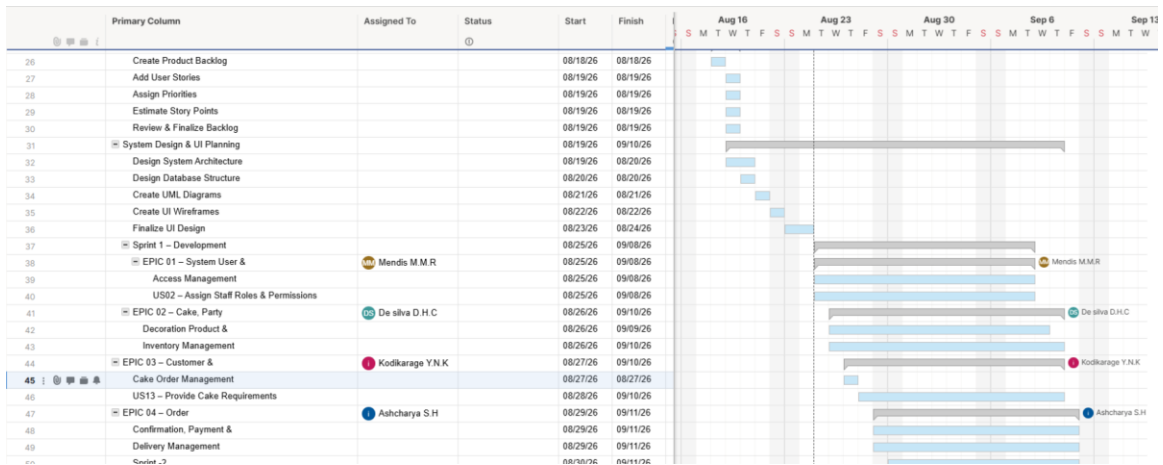
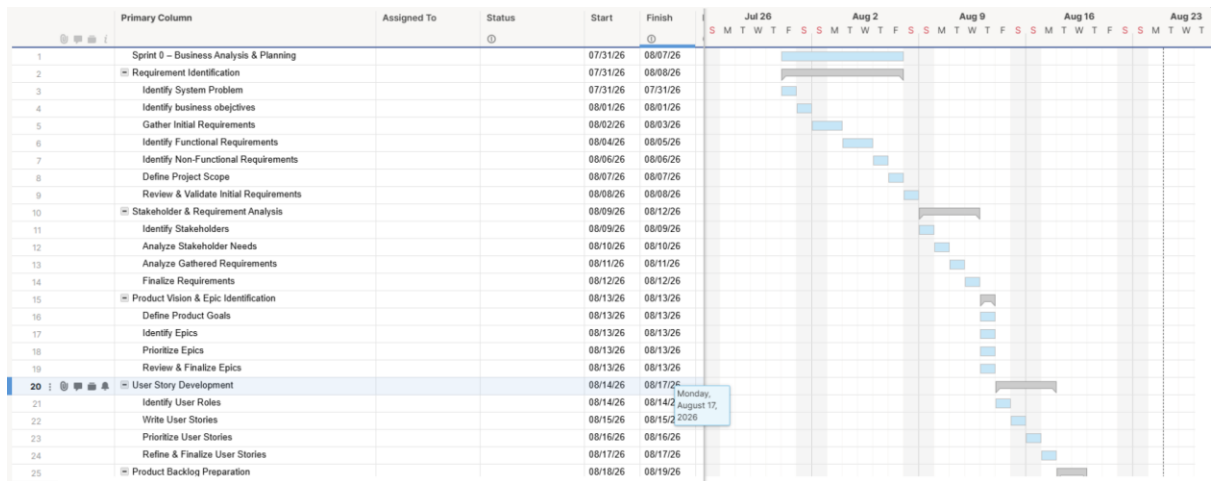
3. Stakeholder Identification

List and categorize key stakeholders:

Stakeholder	Type (Primary / Secondary / Tertiary)	Interest / Concern
Business Owner / Admin	Primary	Oversees the overall business, manages staff access, products and inventory, customer and

		order activities, payment and delivery operations, and makes key business decisions.
Shop Staff	Primary	Handles customer information, manages product and inventory records, processes customer orders, updates order status, and manages payment and delivery or collection activities.
Customer	Primary	Browses available cakes and party decoration products, provides cake requirements, selects products, places orders, and receives order-related services.
System Administrator / Technical Support	Secondary	Maintains the web-based system, manages technical configuration and system access, supports system availability, and resolves technical issues when required.

4. Procedure



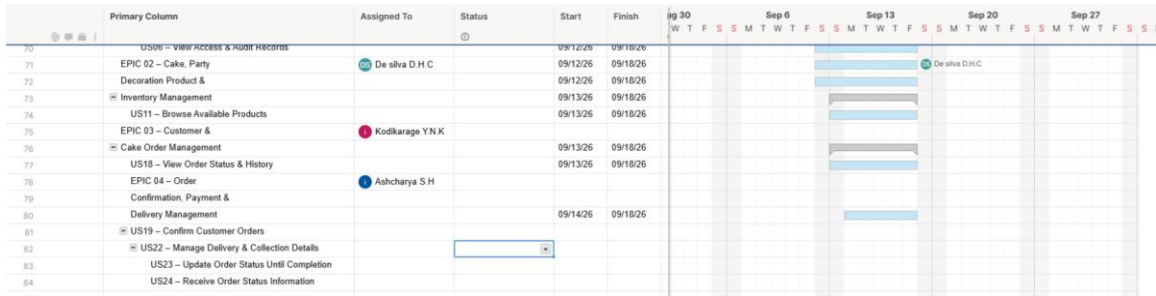


Figure 2 : Preliminary Gantt Chart for the Proposed Project

5. Product Vision Statement

- **For** Devma Cake n' Party and its customers,
- **Who** need a more efficient and organized way to manage and access cake and party decoration services,
- **The product** is a web-based Cake and Party Decoration Management System,
- **Provides** a centralized and reliable platform that improves business efficiency, information visibility, customer service, and the overall management of day-to-day business operations.

6. Epics

Epic ID	Epic Description
EP01 – System User & Access Management	Manage internal staff accounts, user roles, authentication, passwords, and role-based access permissions to ensure that staff members can access only the functions and information authorized for their roles.
EP02 – Cake, Party Decoration Product and Inventory Management	Manage cake and party decoration products and inventory by maintaining product details, categories,

	prices, descriptions, images, availability, and current stock quantities.
EP03 – Customer & Cake Order Management	Manage customer information, cake requirements, party decoration selections, and customer orders while allowing staff to review, update, and maintain customer and order records.
EP04 – Order Confirmation, Payment & Delivery Management	Manage the order process after submission, including order confirmation, payment recording, order status updates, and delivery or collection arrangements until order completion.

7. User Stories

Epic ID	User Story ID	User Story	Stakeholder
EP01	US01	As an Admin, I want to create staff accounts, so that authorized staff members can access the system.	Admin
EP01	US02	As an Admin, I want to assign roles and permissions to staff members, so that they can access only the functions authorized for their roles.	Admin

EP01	US03	As a Staff Member, I want to securely log in to the system, so that I can access the functions available to my role.	Internal Staff
EP01	US04	As a Staff Member, I want to manage my password, so that I can maintain secure access to my account.	Internal Staff
EP01	US05	As an Admin, I want to deactivate staff accounts, so that inactive users cannot access the system.	Admin
EP01	US06	As an Admin, I want to view access and audit records, so that I can monitor system activities and maintain accountability.	Admin
EP02	US07	As an Admin, I want to add new cake and party decoration products, so that available products	Admin

		can be maintained in the system.	
EP02	US08	As a Staff Member, I want to view and search product details, so that I can quickly find the products required for customer orders.	Internal Staff
EP02	US09	As an Admin, I want to update product information, so that prices, descriptions, images, and availability remain accurate.	Admin
EP02	US10	As an Admin, I want to manage inventory quantities, so that current stock levels are accurately maintained.	Admin
EP02	US11	As a Customer, I want to browse available cakes and party decoration products, so that I can	Customer

		select products for my order.	
EP03	US12	As a Customer, I want to create an account and provide my details, so that I can place and manage orders through the system.	Customer
EP03	US13	As a Customer, I want to provide my cake requirements, so that the business can understand and prepare my requested cake.	Customer
EP03	US14	As a Customer, I want to select party decoration products, so that I can include suitable decoration items in my order.	Customer
EP03	US15	As a Customer, I want to submit an order, so that the business can review and process my requirements.	Customer

EP03	US16	As a Staff Member, I want to view customer and order details, so that I can process customer orders accurately.	Internal Staff
EP03	US17	As a Staff Member, I want to update order details and status, so that the current progress of each order is maintained.	Internal Staff
EP03	US18	As a Customer, I want to view my order status and order history, so that I can track my previous and current orders.	Customer
EP04	US19	As a Staff Member, I want to confirm customer orders, so that confirmed orders can proceed to the next stage of processing.	Internal Staff
EP04	US20	As a Staff Member, I want to record payment information, so that payment	Internal Staff

		details are maintained with the relevant order.	
EP04	US21	As a Staff Member, I want to update payment status, so that the payment progress of each order is clear.	Internal Staff
EP04	US22	As a Staff Member, I want to manage delivery or collection details, so that customers can receive their orders through the agreed arrangement.	Internal Staff
EP04	US23	As a Staff Member, I want to update the order status until completion, so that the progress of each order can be tracked.	Internal Staff
EP04	US24	As a Customer, I want to receive order status information, so that I know the progress of my order.	Customer

8. Initial Product Backlog

User Story ID	Remarks	Estimated story points	Estimated hours	Priority (High / Medium / Low)
US01	Create staff accounts	3	6	High
US02	Assign staff roles and permissions	5	10	High
US03	Secure staff login	5	10	High
US04	Manage staff passwords	3	6	Medium
US05	Deactivate staff accounts	3	6	Medium
US06	View access and audit records	5	10	Low
US07	Add cake and party decoration products	5	10	High
US08	View and search product details	3	6	Medium

US09	Update product information	3	6	Medium
US10	Manage inventory quantities	5	10	High
US11	Browse available products	5	10	Medium
US12	Create customer accounts	3	6	High
US13	Provide cake requirements	5	10	High
US14	Select party decoration products	3	6	Medium
US15	Submit customer orders	5	10	High
US16	View customer and order details	3	6	High
US17	Update order details and status	5	10	Medium

US18	View order status and history	3	6	Medium
US19	Confirm customer orders	5	10	High
US20	Record payment information	5	10	High
US21	Update payment status	3	6	Medium
US22	Manage delivery or collection details	5	10	High
US23	Update order status until completion	3	6	Medium
US24	Receive order status information	3	6	Medium

9. Definition of Done (DoD)

A user story will be considered **Done** when:

- The functionality described in the user story has been fully implemented according to the agreed requirements.

- The implemented functionality has been integrated successfully with the relevant system modules.
- The functionality has been tested and works correctly without critical errors.
- Required input validation and appropriate error handling have been implemented.
- The functionality follows the defined security and role-based access requirements where applicable.
- The user interface is usable and consistent with the overall system design.
- Relevant data is correctly stored, retrieved, updated, and managed in the system database.
- The completed functionality has been reviewed and accepted by the team/Product Owner.
- Necessary documentation and supporting information have been updated

10. Sprint 1 Preview

Sprint 1 will concentrate on putting into action the top-priority user stories and the ones that are most important for the business from all four epics that were recognized. The initial scope for Sprint 1 was chosen based on the value for the business the level of priority any dependencies and how ready the work is. Least 50% of the tasks, from every single epic have been added as needed.

Epic	Sprint 1 User Stories	Selected Work (Stories)
EPIC 01 – System User & Access Management	US01, US02, US03	3 / 6 User Stories
EPIC 02 – Cake, Party Decoration Product & Inventory Management	US07, US08, US10	3 / 5 User Stories

EPIC 03 – Customer & Cake Order Management (CRM)	US12, US13, US14, US15	4 / 7 User Stories
EPIC 04 – Order Confirmation, Payment & Delivery Management	US19, US20, US22	3 / 6 User Stories

Selected stories build the core foundation of the system. These selected stories cover staff account management, role-based access, secure authentication, product management, inventory control, customer registration, cake requirements, party decoration choice order submission order confirmation, payment recording and delivery or collection management. The Sprint 1 scope is a first draft. It may change after the Sprint 0 review, backlog refinement and team capacity assessment.

11. Software and Hardware Requirements

Software Requirements

Software / Tool	Purpose
HTML5, CSS3, JavaScript	Develop the web-based user interface and client-side functionality.
React.js	Develop the interactive web application user interface.
Node.js & Express.js	Develop the server-side application and RESTful APIs.
MySQL	Store and manage staff, product, inventory, customer, order, payment, and delivery information.

Visual Studio Code	Main IDE for developing and maintaining the system source code.
Git & GitHub	Version control and collaborative source-code management.
Figma	Design user interfaces, wireframes, and prototypes.
Draw.io	Create system architecture diagrams, UML diagrams, flowcharts, and other project diagrams.
Postman	Test and verify RESTful APIs and backend functionality.
Web Browser (Google Chrome / Microsoft Edge)	Run, test, and debug the web-based system.

Hardware Requirements

Hardware	Purpose / Requirement
Developer Laptop / PC	Used by team members for system development, testing, documentation, and project management.
Processor	Intel Core i5 / AMD Ryzen 5 or equivalent and above.
RAM	Minimum 8 GB; 16 GB recommended for development.
Storage	Minimum 256 GB SSD; 512 GB SSD recommended.
Internet Connection	Required for collaboration, GitHub, package installation, API testing, and cloud-based services where applicable.

Smartphone / Tablet	Used for responsive design and mobile-browser testing.
Deployment Server / Cloud Hosting	Used to host the web application and database when the system is deployed.

12.Budget

Selected stories build the core foundation of the system. I believe these selected stories cover staff account management, role-based access, secure authentication, product management, inventory control, customer registration, cake requirements, party decoration choice order submission order confirmation, payment recording and delivery or collection management. The Sprint 1 scope is a draft. It may change after the Sprint 0 review, backlog refinement and team capacity assessment.

Cost Item	Description	Estimated Cost (LKR)
Development Software	Free development tools such as Visual Studio Code and Git	0
Database Software	Free/open-source database software	0
Design and Diagram Tools	Figma and Draw.io free versions	0
Development Hardware	Existing team laptops/PCs	0
Testing Devices	Existing smartphones and computers	0
Internet Usage	Existing internet connection used by team members	0

Hosting / Domain	Not required for the academic demonstration	0
Third-Party APIs	No paid third-party APIs required	0
Software Licenses	No additional paid software licenses required	0
Total Estimated Cost		0

Budget Assumption

The proposed Devma Cake n' Party Management System is being built for evaluation and demonstration. Therefore the Devma Cake n' Party Management System does not need hosting, domain registration, paid APIs or extra hardware. The team will use existing hardware and free software tools. Consequently the estimated extra cost, for the project is LKR 0.

13. References

- [1] Atlassian, “How to write SMART goals,” Atlassian. [Online]. Available: <https://www.atlassian.com/blog/teamwork/how-to-write-smart-goals>. [Accessed: Aug. 25, 2026].
- [2] Atlassian, “SMART goals template,” Atlassian Confluence. [Online]. Available: <https://www.atlassian.com/software/confluence/templates/smart-goals>. [Accessed: Aug. 25, 2026].
- [3] diagrams.net, “diagrams.net documentation.” [Online]. Available: <https://www.drawio.com/doc/>. [Accessed: Aug. 25, 2026].
- [4] Figma, “Figma Help Center.” [Online]. Available: <https://help.figma.com/>. [Accessed: Aug. 25, 2026].

14.Appendix

The following supporting materials are included in the appendix to give information and proof about the Devma Cake n' Party Management System that is being proposed.

Appendix A – Project Charter

The project charter that has been approved which includes the project background the range of work the people involved the goals, the team details, the things that are assumed to be true the limits that're in place and other details about the project.

Appendix B – Enterprise Architecture / Sprint 0 Supporting Documents

Documents from Sprint 0 that were made during the Enterprise Architecture module, such, as analysis of the people involved how they are affected, how the processes work and other analysis that supports the project.